Reproducing and Stress-Testing PerSAM for One-Shot Personalized Segmentation

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1 Introduction

The Segment Anything Model (SAM) is a strong promptable segmentation model, but it still requires a user prompt for each target image. This is limiting when a user wants to repeatedly segment the same personalized object across many images. Zhang et al. address this in *Personalize Segment Anything Model with One Shot* by proposing PerSAM, which customizes SAM from one reference image and one reference mask [1]. PerSAM extracts a target representation from the reference mask, localizes similar regions in new images, and guides SAM to segment the user-specified object. The paper also proposes PerSAM-F, which freezes SAM and tunes only two mask-scale weights to reduce part-vs-whole mask ambiguity.

Our project reproduces the main PerSeg personalized segmentation result and adds a robustness experiment. We ask whether PerSAM still works when the user-provided reference mask is imperfect, since real users may provide rough, noisy, or misaligned masks.

2 Chosen Result

We reproduce the PerSeg personalized object segmentation result, which directly tests the paper's central claim: SAM can be personalized to segment a specific visual concept from a single reference image-mask pair. We focus on PerSAM and PerSAM-F instead of the video or DreamBooth extensions because PerSeg is the cleanest benchmark for one-shot personalized segmentation.

The paper reports that PerSAM reaches about 89.3 mIoU on PerSeg, while PerSAM-F improves to about 95.3 mIoU through lightweight scale-aware fine-tuning [1]. We reproduce this result using the official codebase, PerSeg dataset, SAM ViT-H checkpoint, and the provided mIoU evaluation script.

3 Methodology

We used the official PerSAM implementation. Our pipeline was: download and organize PerSeg into `Images/` and `Annotations/`; run PerSAM and PerSAM-F on Google Colab with an L4 GPU and SAM ViT-H; evaluate predicted masks using `eval_miou.py`. PerSAM is training-free: for each object category, it uses the reference image and mask indexed by `00`, computes a similarity-based location prior, selects positive/negative prompt locations, and guides SAM through target information. PerSAM-F adds a small tuning step while freezing SAM.

For our independent experiment, we perturbed only the reference mask and kept all evaluation ground-truth masks unchanged. We tested erosion, dilation, spatial shift, and random pixel noise. This isolates how imperfect one-shot supervision affects PerSAM. A key implementation issue was mask encoding: some PerSeg foreground masks use value 38 rather than 255. A naive threshold at 127 erased the object, so our script binarized masks using `mask > 0`.

4 Results & Analysis

Table 1 shows that our reproduction matches the main trend of the paper. PerSAM achieves 89.32 mIoU and 92.19 mAcc. PerSAM-F improves to 95.18 mIoU and 95.57 mAcc, a +5.86 mIoU gain.

The improvement is driven by hard PerSAM failure cases: `can` improves from 38.91 to 97.49 IoU, `rc_car` from 39.30 to 96.12, `teapot` from 40.02 to 96.93, and `robot_toy` from 60.55 to 96.66. This supports the claim that PerSAM-F helps when PerSAM chooses an incorrect mask scale. However, PerSAM-F is not uniformly better. For `colorful_teapot`, IoU drops from 96.27 to 84.48. This object is visually distinctive and already easy for

Table 1: Main reproduction results on PerSeg.

Method	mIoU	mAcc
PerSAM	89.32	92.19
PerSAM-F	95.18	95.57
Improvement	+5.86	+3.38

PerSAM; additional scale tuning may over-correct around thin structures such as the handle and spout.

Table 2 summarizes our robustness experiment. Erosion and dilation cause only small drops, and sparse random noise has no measurable effect. In contrast, shifting the reference mask sharply decreases mIoU from 89.32 to 73.84. This suggests that PerSAM is robust to imperfect boundaries but sensitive to spatial misalignment. Since the target embedding is extracted from the masked reference region, a shifted mask can contaminate the representation with background or wrong-object features.

Table 2: PerSAM robustness to imperfect reference masks.

Reference Mask	mIoU	mAcc	Δ mIoU
Original	89.32	92.19	–
Erode	88.45	92.18	-0.87
Dilate	87.33	90.23	-1.99
Shift	73.84	82.28	-15.48
Noise	89.32	92.19	0.00

5 Reflections

This project showed that reproducing a foundation-model paper is not only about running the main script. Much of the work involved setting up data paths, checkpoints, evaluation scripts, and mask formats correctly. The mask-value bug was especially important: using `mask > 127` destroyed masks whose foreground value was 38, while `mask > 0` fixed the perturbation pipeline.

The main takeaway is that PerSAM efficiently turns SAM into a one-shot personalized segmenter. PerSAM-F improves many hard cases by resolving scale ambiguity, but it can hurt easy objects where PerSAM already works well. Our robustness experiment adds a practical lesson: boundary quality matters less than reference-mask alignment. Future work could test multiple reference examples, automatic reference-mask correction, or stronger real-world user annotation noise.

6 References

References

[1] Renrui Zhang, Zhengkai Jiang, Ziyu Guo, Shilin Yan, Junting Pan, Xianzheng Ma, Hao Dong, Peng Gao, and Hongsheng Li. *Personalize Segment Anything Model with One Shot*. arXiv:2305.03048, 2023.

[2] Alexander Kirillov et al. *Segment Anything*. ICCV, 2023.

[3] PerSAM official repository. `https://github.com/ZrrSkywalker/Personalize-SAM`.

[4] Segment Anything official repository. `https://github.com/facebookresearch/segment-anything`.